$$= 9\left(-\cos\left(\frac{\pi}{6}\right)-i\sin\left(\frac{\pi}{6}\right)\right)$$

$$= 9\left(-\frac{\sqrt{3}}{2}-\frac{i}{2}\right)=-\frac{9\sqrt{3}}{2}-\frac{9i}{2}\text{ , Which is in rectangular form.}$$

Example 2.26

Find the quotient $\dfrac{2\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)}{4\left(\cos\left(\frac{-3\pi}{2}\right)+i\sin\left(\frac{-3\pi}{2}\right)\right)}$ in rectangular form.

Solution

$$\frac{2\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)}{4\left(\cos\left(\frac{-3\pi}{2}\right)+i\sin\left(\frac{-3\pi}{2}\right)\right)}$$

$$= \frac{1}{2}\left(\cos\left(\frac{9\pi}{4}-\left(\frac{-3\pi}{2}\right)\right)+i\sin\left(\frac{9\pi}{4}-\left(\frac{-3\pi}{2}\right)\right)\right)$$

$$= \frac{1}{2}\left(\cos\left(\frac{9\pi}{4}+\frac{3\pi}{2}\right)+i\sin\left(\frac{9\pi}{4}+\frac{3\pi}{2}\right)\right)$$

$$= \frac{1}{2}\left(\cos\left(\frac{15\pi}{4}\right)+i\sin\left(\frac{15\pi}{4}\right)\right)=\frac{1}{2}\left(\cos\left(4\pi-\frac{\pi}{4}\right)+i\sin\left(4\pi-\frac{\pi}{4}\right)\right)$$

$$= \frac{1}{2}\left(\cos\left(\frac{\pi}{4}\right)-i\sin\left(\frac{\pi}{4}\right)\right)=\frac{1}{2}\left(\frac{1}{\sqrt{2}}-i\frac{1}{\sqrt{2}}\right)$$

$\dfrac{2\left(\cos\frac{9\pi}{4}+i\sin\frac{9\pi}{4}\right)}{4\left(\cos\left(\frac{-3\pi}{2}\right)+i\sin\left(\frac{-3\pi}{2}\right)\right)} = \dfrac{1}{2\sqrt{2}}-i\dfrac{1}{2\sqrt{2}}=\dfrac{\sqrt{2}}{4}-i\dfrac{\sqrt{2}}{4}$. Which is in rectangular form.

Example 2.27

If $z=x+iy$ and $\arg\left(\dfrac{z-1}{z+1}\right)=\dfrac{\pi}{2}$, show that $x^2+y^2=1$.

Solution

$$\text{Now, } \frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1}=\frac{(x-1)+iy}{(x+1)+iy} = \frac{\left[(x-1)+iy\right]\left[(x+1)-iy\right]}{\left[(x+1)+iy\right]\left[(x+1)-iy\right]}$$

$$\Rightarrow \frac{z-1}{z+1} = \frac{(x^2+y^2-1)+i(2y)}{(x+1)^2+y^2}.$$

$$\text{Since, } \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2} \Rightarrow \tan^{-1}\left(\frac{2y}{x^2+y^2-1}\right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{2y}{x^2+y^2-1} = \tan\frac{\pi}{2} \Rightarrow x^2+y^2-1=0$$

$$\Rightarrow x^2+y^2 = 1.$$

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